

SemiAuto Classification Report

Dataset: JEE_DROPOUT_AFTER_CLASS_12

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Project Flow



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1. Data Ingestion

Initial dataset analysis and characteristics.

Dataset Overview

Dataset: JEE_Dropout_After_Class_12

Train samples: 4000, **Test samples:** 1000

Target column: dropout

Column Types

Numerical Columns:

jee_main_score, jee_advanced_score, mock_test_score_avg, class_12_percent, attempt_count, daily_study_hours, dropout

Categorical Columns:

school_board, coaching_institute, family_income, parent_education, location_type, peer_pressure_level, mental_health_issues, admission_taken

Columns with Nulls:

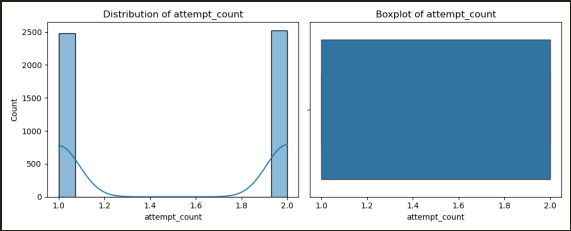
coaching_institute

Columns with Outliers:

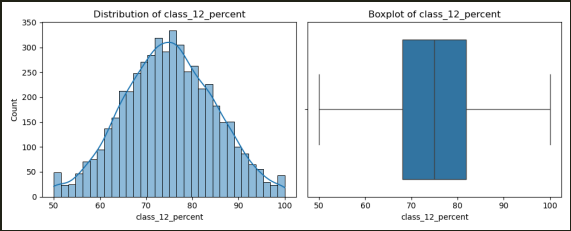
jee_main_score, jee_advanced_score, mock_test_score_avg, dropout

Feature Distributions

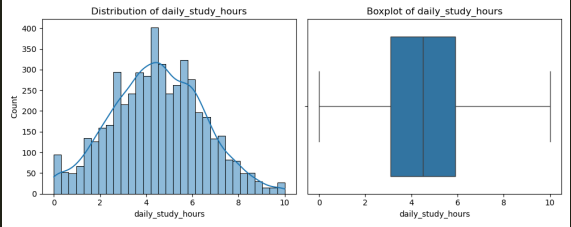
attempt_count



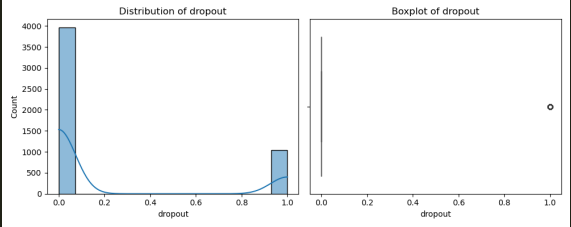
class_12_percent



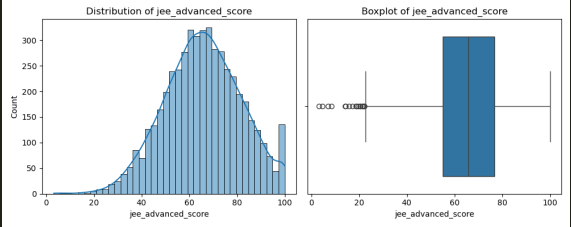
daily_study_hours



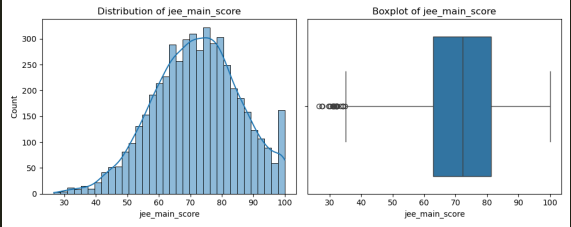
dropout



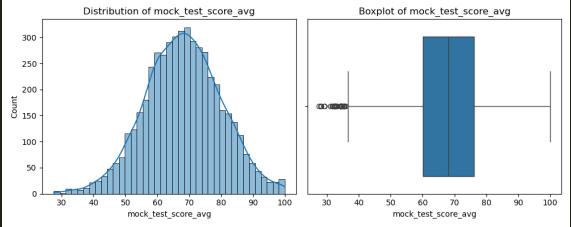
jee_advanced_score



jee_main_score



mock_test_score_avg



2. Data Preprocessing

Data cleaning and preparation steps.

3. Feature Engineering

Feature selection and engineering steps.

Feature Engineering Config

Feature Selection Method:

None

4. Model Building

Model training configuration.

Model Selection

Selected Model:

CatBoost_2

5. Model Evaluation

Model performance metrics.

Classification Metrics

Metric	Value
Accuracy	0.9970
Precision	0.9951
Recall	0.9903
F1 Score	0.9927
ROC AUC	0.9977

6. Model Optimization

Hyperparameter tuning results.

Error: Invalid JSON format in hyperparameters file

7. Final Evaluation Results

Optimized model performance.

Optimized Metrics

Metric	Original	Optimized	Change
Accuracy	0.997	0.997	0.0000
Precision	0.9951456310679612	0.9951456310679612	0.0000
Recall	0.9903381642512077	0.9903381642512077	0.0000
F1 Score	0.9927360774818402	0.9927360774818402	0.0000
ROC AUC	0.99768505826952	0.9987816096155369	0.0011